

WanderBricks Bookings — EDA

A simple student-style exploratory analysis using SQL

1. Introduction

I worked on the WanderBricks bookings dataset to understand the basic booking pattern, booking status, revenue, stay duration, guest count and property performance. I used SQL queries to explore the data and then noted the results that would be useful for a Power BI dashboard.

2. First look at the data

The first step was to check the available tables and inspect the bookings table. I used SHOW TABLES, selected a few records and described the table before starting the analysis.

```
show tables;

select * from bookings limit 10;

describe bookings;
```

3. Size and shape of the dataset

I checked the total number of booking records and also checked booking IDs and unique users. This helps in understanding how large the booking activity is and whether many bookings belong to the same users.

```
select count(*) from bookings;

select count(booking_id) from bookings;

select count(distinct(user_id)) from bookings;
```

4. Booking status

Next, I checked the different booking statuses and counted how many bookings were present in each status. This is useful because a booking that is cancelled should not be treated the same as a completed booking.

```
select status, count(booking_id)
from bookings
group by status;

select distinct status from bookings;

select status, count(*) as booking_count
from bookings
group by status
order by booking_count desc;
```

5. Actual / successful bookings

I also checked confirmed and completed bookings separately. For revenue and property performance, completed bookings are especially important because they represent stays that were actually completed.

```
select count(property_id)
from bookings
where status = 'confirmed' or status = 'completed';

select status, count(property_id)
from bookings
```

```
group by status
having status = 'confirmed' or status = 'completed';
```

6. Revenue by property type

To compare property types, I joined properties with bookings and calculated completed-booking revenue. The intended idea is to see which property types generate more revenue.

```
create or replace temp view properties_type_view as
select p.property_type, b.*
from properties p
join bookings b on b.property_id = b.property_id;

select property_type, status,
       round(sum(total_amount),2) as total_revenue
from properties_type_view
group by property_type, status
having status = 'completed';
```

Note: the join condition in the original query is written as `b.property_id = b.property_id`. That condition is always true for non-null property IDs and can create an incorrect many-to-many result. For a final dashboard, I would use the intended relationship, `p.property_id = b.property_id`.

7. Booking period and seasonal analysis

I checked the earliest and latest booking creation dates to understand the time period covered by the dataset. After that, booking counts by status can be used for trend analysis in Power BI.

```
select min(created_at) as first_booking,
       max(created_at) as latest_booking
from bookings;

select status, count(*) as booking_count
from bookings
group by status
order by booking_count desc;
```

8. Stay duration

Stay duration can be calculated by finding the difference between check-out and check-in. This can help identify whether customers usually book short stays or longer stays.

```
select datediff(check_out, check_in)
from bookings;
```

9. Average guests and booking amount

I calculated the average number of guests for each property type. I also tried to calculate average booking amount by property type. The second query should be interpreted as average total_amount rather than average guests, because the selected field is total_amount.

```
select property_type,
       round(avg(guests_count),0) as avg_guests
from properties_type_view
group by property_type;

select property_type,
       round(avg(total_amount),0) as avg_guests
from properties_type_view
group by property_type;
```

10. Repeat booking behaviour

I counted bookings for each user and sorted the result from highest to lowest. This gives a simple way to identify users who book repeatedly and could be useful for customer segmentation.

```
select user_id, count(*) as total_count
from bookings
group by user_id
order by total_count desc
limit 50;
```

11. Property performance

Finally, I calculated the average booking amount for each property. The query compares each property's average with the overall average booking amount and assigns a simple performance label.

```
select property_id,
       avg(total_amount) as avgtotalAmount,
       case
         when avg(total_amount) >
              (select avg(total_amount) from bookings)
         then "Good"
         else "bad"
       end as performance
from bookings
group by property_id;
```

12. Dashboard ideas

KPI	What I would show
Total bookings	Total booking records
Unique users	Distinct user count
Completed bookings	Bookings with completed status
Revenue	Sum of total_amount for completed bookings
Booking status	Count by status
Property type	Revenue and average guests by property type
Stay duration	Average / distribution of check-in to check-out days
Repeat users	Users with multiple bookings
Property performance	Average booking amount compared with overall average

13. Conclusion

This analysis gave me a basic understanding of the WanderBricks booking data. The main areas I would use in the final Power BI dashboard are booking status, completed revenue, property type, guest count, stay duration, repeat users and property-level performance. Before using the results for a final report, I would fix the property-to-booking join condition and rename the average amount column correctly.

Source: WanderBricks bookings EDA SQL file supplied for this project.